

5. Project Development

Project Name

ComicCraft - AI Comic Story Creator using Gemini Models

1. Introduction

Project Development is the implementation phase of ComicCraft. In this phase, the frontend, backend, Gemini AI integration, and comic preview features are developed and connected together.

2. Development Technologies

- Frontend: HTML, CSS, JavaScript
- Backend: Python Flask
- AI Model: Google Gemini
- Code Editor: Visual Studio Code
- Version Control: Git and GitHub
- Web Browser: Google Chrome

3. Development Modules

3.1 Frontend Development

The frontend provides an easy-to-use interface for users.

Main features include:

- Story idea input
- Character details input
- Generate button
- Comic preview
- Simple navigation

3.2 Backend Development

The backend is developed using Flask.

The backend is responsible for:

- Receiving user input
- Processing requests

- Communicating with Gemini AI
- Handling API responses
- Sending generated content to the frontend

3.3 Gemini AI Integration

Google Gemini is used to generate creative comic content based on the user's input.

The AI can generate:

- Story ideas
- Characters
- Scenes
- Dialogues
- Story descriptions

3.4 Comic Preview

The generated content is presented in a comic-style preview.

The preview contains:

- Character information
 - Scene descriptions
 - Dialogues
 - Story sequence
- #### 4. Development Workflow

User enters story idea | v Frontend sends request | v Flask Backend | v Gemini AI | v
Generated Story Content | v Backend processes response | v Comic Preview

5. Project Structure

```
ComicCraft AI/ | ├── app.py |── routes.py |── requirements.txt |── templates/ |
|── index.html |── comic_preview.html |── static/ |── style.css |──
1_Brainstorming_and_Ideation/ |── README.md |── 2_Requirement_Analysis/ |──
README.md |── 3_Project_Design/ |── README.md |── 4_Project_Planning/ |──
README.md |── 5_Project_Development/ |── README.md
```

6. Development Steps

Step 1: Create the Project

Create the ComicCraft project folder and organize the required files and folders.

Step 2: Develop the Frontend

Create HTML and CSS files for the user interface.

Step 3: Develop the Flask Backend

Create Flask routes to handle user requests and application logic.

Step 4: Integrate Gemini AI

Connect the application with the Gemini API to generate comic story content.

Step 5: Connect Frontend and Backend

Connect the user interface with the Flask backend.

Step 6: Create Comic Preview

Display the generated story, characters, scenes, and dialogues in the comic preview page.

Step 7: Test the Application

Test the complete workflow and fix errors.

7. API Integration

The Gemini API is used to process user prompts and generate AI-based comic story content.

The basic process is:

User Prompt ↓ Gemini API Request ↓ AI Processing ↓ Generated Response ↓ Flask Backend ↓ Comic Preview

8. Error Handling

The application should handle common errors such as:

- Empty user input
- Invalid requests

- API connection errors
- API response errors
- Server errors

Appropriate error messages should be displayed to the user.

9. Testing During Development

The following features should be tested:

Feature| Test Story Input| Check whether user input is accepted Character Input| Check character details Gemini API| Check AI response Story Generation| Check generated content Comic Preview| Check displayed content Navigation| Check page-to-page navigation Error Handling| Check invalid inputs

10. Development Status

Component| Status Frontend| In Progress Flask Backend| In Progress Gemini Integration| In Progress Story Generation| In Progress Comic Preview| In Progress Testing| Planned Documentation| In Progress

11. Expected Output

The completed application should allow users to enter a story idea and generate an AI-powered comic story containing characters, scenes, and dialogues.

12. Conclusion

The Project Development phase converts the planned design into a functional ComicCraft application. The frontend, Flask backend, and Gemini AI are integrated to provide an AI-powered comic story creation experience.